

HSCoDeCOMP: A Realistic and Expert-level Benchmark for Deep Search Agents in Hierarchical Rule Application

HSCoDeCOMP: 一个用于分层规则应用中深度搜索代理的现实且专家级基准

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Alibaba International Digital Commerce

阿里巴巴国际数字贸易

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Abstract

摘要

Effective deep search agents must not only access open-domain and domain-specific knowledge but also apply complex rules—such as legal clauses, medical manuals and tariff rules. These rules often feature vague boundaries and implicit logic relationships, making precise application challenging for agents. However, this critical capability is largely overlooked by current agent benchmarks. To fill this gap, we introduce HSCoDeCOMP, the first realistic, expert-level e-commerce benchmark designed to evaluate deep search agents in hierarchical rule application. In this task, the deep reasoning process of agents is guided by these rules to predict 10-digit Harmonized System Code (HSCoDe) of products with noisy but realistic descriptions. These codes, established by the World Customs Organization, are vital for global supply chain efficiency. Built from real-world data collected from large-scale e-commerce platforms, our proposed HSCoDeCOMP comprises 632 product entries spanning diverse product categories, with these HSCodes annotated by several human experts. Extensive experimental results on several state-of-the-art LLMs, open-source, and closed-source agents reveal a huge performance gap: best agent achieves only 46.8% 10-digit accuracy, far below human experts at 95.0%. Besides, detailed analysis demonstrates the challenges of hierarchical rule application, and test-time scaling fails to improve performance further.

有效的深度搜索代理不仅需要访问开放领域和特定领域的知识, 还需要应用复杂的规则, 例如法律条款、医疗手册和关税规则。这些规则通常具有模糊的边界和隐含的逻辑关系, 使得代理在精确应用时面临挑战。然而, 这种关键能力在当前的代理基准测试中被严重忽视。为了填补这一空白, 我们引入了 HSCoDeCOMP, 这是首个真实、专家级的电子商务基准测试, 旨在评估深度搜索代理在分层规则应用方面的能力。在这个任务中, 代理的深度推理过程由这些规则引导, 以预测具有噪声但现实的产品描述的 10 位《协调制度》编码 (HSCoDe)。这些编码由世界海关组织制定, 对全球供应链效率至关重要。HSCoDeCOMP 是基于从大规模电子商务平台收集的现实数据构建的, 包含 632 个产品条目, 涵盖多种产品类别, 并由多位人类专家标注了这些 HSCoDe。在多个最先进的大型语言模型 (LLMs)、开源和闭源代理上的广泛实验结果揭示了巨大的性能差距: 最佳代理仅达到 46.8% 位数字的准确性, 远低于人类专家的 95.0%。此外, 详细分析展示了分层规则应用的挑战, 且测试时的扩展并未进一步提高性能。



Figure 1: images/7af01b72bb70b6461b92d2cfc2cf0408572eaa53ecb416856fd69b7c36629b11.jpg

<https://github.com/AIDC-AI/Marco-Search-Agent>

<https://github.com/AIDC-AI/Marco-Search-Agent>



Figure 2: images/86dea25cd1c83ecd0a7639ef65efb1ea089150ef37ab08f0a3af07153415d336.jpg

<https://huggingface.co/datasets/AIDC-AI/HSCodeComp>

<https://huggingface.co/datasets/AIDC-AI/HSCodeComp>

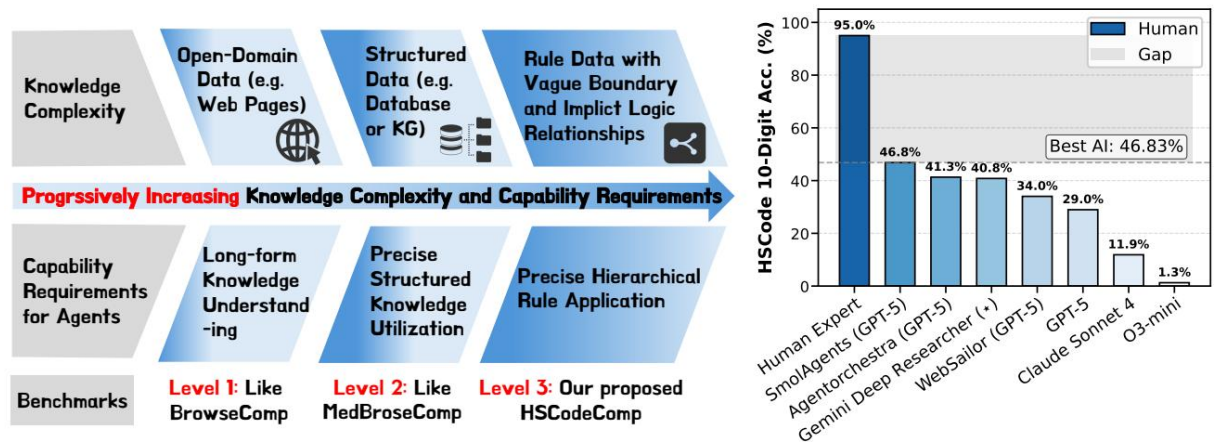


Figure 1. Left: Recent benchmarks reveal the increasing knowledge complexity and capability requirements for agents. Right: 10-digit HSCode accuracy of state-of-the-art baseline largely lags behind human experts ($46.8\% < 95.0\%$), proving the challenges of hierarchical rule application. The closed-source agent (★) is evaluated on the subset due to API unavailability.

图 1. 左：近期基准测试揭示了代理所需的知识复杂性和能力要求的不断提高。右：当前最先进的基线模型在 10 位 HS 编码上的准确性远低于专家水平 ($46.8\% < 95.0\%$)，证明了分层规则应用的挑战性。由于 API 不可用，封闭源代理 (★) 在子集上进行了评估。

1. Introduction

1. 引言

Deep search agents have demonstrated significant value in solving complex real-world problems, where robust external knowledge utilization constitutes a critical capability [Wu et al., 2025, Tao et al., 2025, Li et al., 2025b]. To evaluate this capability, numerous established benchmarks are proposed to assess agents in utilizing open-domain data (e.g., GAIA [Mialon et al., 2023b] and BrowseComp [Wei et al., 2025]) and domain-specific data (e.g., WebMall [Peeters et al., 2025a], FinSearchComp [Hu et al., 2025a] and MedBrowseComp [Yu et al., 2025b]).

深度搜索代理在解决复杂的现实世界问题中展现出显著的价值，其中稳健地利用外部知识是一项关键能力

[Wu 等, 2025, Tao 等, 2025, Li 等, 2025b]。为了评估这一能力, 提出了许多已建立的基准测试, 用于评估代理利用开放领域数据 (如 GAIA [Mialon 等, 2023b] 和 BrowseComp [Wei 等, 2025]) 和领域特定数据 (如 WebMall [Peeters 等, 2025a]、FinSearchComp [Hu 等, 2025a] 和 MedBrowseComp [Yu 等, 2025b]) 的能力。

Beyond open-domain and domain-specific data, agents also need to effectively apply rules that encode human expert knowledge, particularly in scenarios like law, medical and e-commerce [Li et al., 2025a, Chen et al., 2025b, Yao et al., 2022, Chollet et al., 2025]. For instance, legal case adjudication require interpreting abstract legal provisions, and accurate e-commerce product classification in depends on tariff rules [Grainger, 2024]. Previous works have defined rule application as using specific logical rules with supporting facts to derive conclusions [Wang et al., 2024, Servantez et al., 2024]. In contrast, we define it as a core capability for deep search agents, where human-written rules are systematically applied to guide complex reasoning and decision-making [Sadowski and Chudziak, 2025]. Building on this observation, we categorize knowledge data for deep search agents into three levels (Figure 1, left), with increasing knowledge complexity: (1) Level 1: Open-domain Data - Tests understanding and deep reasoning abilities of agents on long-form web content. Established benchmarks include GAIA [Mialon et al., 2023b] and BrowseComp [Wei et al., 2025]; (2) Level 2: Structured Data - Assesses agents to precisely utilize structured data such as databases and knowledge graphs, as seen in domain-specific benchmarks like WebMall [Peeters et al., 2025a], MedBrowseComp [Chen et al., 2025b] and FinSearchComp [Hu et al., 2025a]; (3) Level 3: Rule Data - Evaluates agents to apply complex and abstract rules [Chollet et al., 2025]. This level presents two key challenges: (a) making accurate decisions when rules contain vague natural language descriptions [Sadowski and Chudziak, 2025]; and (b) reasoning about logical dependencies among rules, such as exception clauses and cross-category relationships [Guha et al., 2023]. Despite the importance of rule application in real-world scenarios, current agent benchmarks largely overlook its evaluation.

除了开放领域和特定领域的的数据, 智能体还需要有效地应用编码人类专家知识的规则, 尤其是在法律、医疗和电子商务等场景中 [Li 等人, 2025a, Chen 等人, 2025b, Yao 等人, 2022, Chollet 等人, 2025]。例如, 法律案件的裁决需要解释抽象的法律条文, 而电子商务产品分类的准确性则依赖于关税规则 [Grainger, 2024]。先前的研究将规则应用定义为使用特定逻辑规则和支持事实来推导结论 [Wang 等人, 2024, Servantez 等人, 2024]。相比之下, 我们将规则应用定义为深度搜索智能体的核心能力, 其中人类编写的规则被系统地应用于指导复杂的推理和决策 [Sadowski 和 Chudziak, 2025]。基于这一观察, 我们将深度搜索智能体的知识数据分为三个层级 (图 1, 左侧), 随着知识复杂性的增加: (1) 第一层: 开放领域数据 - 测试智能体在长文本网络内容上的理解和深度推理能力。已建立的基准包括 GAIA [Mialon 等人, 2023b] 和 BrowseComp [Wei 等人, 2025]; (2) 第二层: 结构化数据 - 评估智能体精确利用结构化数据 (如数据库和知识图谱) 的能力, 如在特定领域基准 WebMall [Peeters 等人, 2025a]、MedBrowseComp [Chen 等人, 2025b] 和 FinSearchComp [Hu 等人, 2025a] 中所见; (3) 第三层: 规则数据 - 评估智能体应用复杂和抽象规则的能力 [Chollet 等人, 2025]。这一层级提出了两个关键挑战: (a) 当规则包含模糊的自然语言描述时做出准确决策 [Sadowski 和 Chudziak, 2025]; 以及 (b) 推理规则之间的逻辑依赖关系, 例如例外条款和跨类别关系 [Guha 等人, 2023]。尽管规则应用在现实场景中非常重要, 但当前的智能体基准测试在很大程度上忽略了对其的评估。

To fill this gap, we introduce HSCODECOMP (short for the Harmonized System Code (HSCode) Competition), the first realistic, expert-level e-commerce benchmark designed to evaluate agents in predicting complete 10-digit Harmonized System Code (HSCode) of the product, using hierarchical rules (e.g., eWTP tariff rules¹). HS Codes organize products through a hierarchical structure spanning over 5,000 distinct codes across multiple classification levels, representing the global standard for classifying traded international goods, established by the World Customs Organization and implemented across more than 200 countries for customs clearance and tariff determination [Grainger, 2024, Nath et al., 2025]. Built from the data of the large-scale e-commerce platforms, our proposed HSCODECOMP comprises 632 carefully curated product entries, encompassing 27 unique HS chapters and 32 distinct first-level categories. These HS Codes have been rigorously annotated by multiple e-commerce domain experts, ensuring that HSCODECOMP is expert-level. Accurately predicting the exact 10-digit HSCode presents significant challenges: agents must perform multi-hop hierarchical reasoning with complex tariff rules while processing noisy but realistic product descriptions that often contain abbreviations, language variations, or incomplete information.

为了解决这一问题，我们引入了 HSCODECOMP（即《协调制度代码》(HSCode) 竞赛的缩写），这是首个真实、专家级的电子商务基准测试，旨在评估智能体在使用分层规则（例如 eWTP 关税规则¹）预测产品完整的 10 位《协调制度代码》(HSCode) 方面的能力。HSCode 通过一个层级结构对产品进行组织，该结构覆盖了多个分类层级上的超过 5000 个不同的代码，代表了世界海关组织制定的用于分类国际贸易商品的全球标准，并在超过 200 个国家实施，用于海关清关和关税确定 [Grainger, 2024, Nath et al., 2025]。HSCODECOMP 基于大型电商平台的数据构建，包含 632 个经过精心整理的产品条目，涵盖 27 个独特的 HS 章节和 32 个不同的第一级类别。这些 HSCode 由多位电子商务领域专家进行了严格标注，确保 HSCODECOMP 具有专家级水平。准确预测确切的 10 位 HSCode 具有重大挑战：智能体必须在处理包含缩写、语言变化或信息不完整的现实产品描述的同时，进行多步分层推理，并应对复杂的关税规则。

Extensive experiments on the state-of-the-art baselines, including 14 advanced foundation models, 6 advanced open-source agent systems and 3 closed-source agent systems, demonstrate that HSCode prediction task remains a substantial challenge for current AI approaches. As shown in the Figure 1 (right), even the best-performing system (SmolAgent [Roucher et al., 2025] with GPT-5) achieves

在对当前最先进的基线模型进行了广泛的实验，包括 14 个先进的基础模型、6 个先进的开源代理系统和 3 个闭源代理系统，结果表明，HSCode 预测任务仍然是当前 AI 方法面临的重大挑战。如图 1（右）所示，即使表现最好的系统（配备 GPT-5 的 SmolAgent [Roucher 等，2025]）也仅能实现

only 46.8% accuracy, substantially below the 95.0% accuracy attained by human experts. Further detailed analysis reveals that existing agent systems lack critical capabilities required for this complex hierarchical rule applications. Notably, test-time scaling approach—which has proven effective in other reasoning tasks [Guo et al., 2025, Liu et al., 2025]—fail to improve performance on HSCODECOMP. These observations demonstrate the challenging nature of our proposed HSCODECOMP, highlighting the need for more effective designs of agent systems. To facilitate future research, we will publicly release codes and the benchmark dataset of HSCODECOMP.

仅有 46.8% 的准确率，远低于人类专家达到的 95.0% 的准确率。进一步详细分析表明，现有的代理系统缺乏执行这种复杂分层规则应用所需的关键能力。值得注意的是，已被证明在其他推理任务中有效的测试时扩展方法 [Guo 等人，2025, Liu 等人，2025] 在 HSCODECOMP 上未能提升性能。这些观察结果展示了我们提出的 HSCODECOMP 的挑战性，突显了需要更有效的代理系统设计的必要性。为了促进未来的研究，我们将公开发布 HSCODECOMP 的代码和基准数据集。

2. Related Works

2. 相关工作

2.1. Previous Works in HSCode Prediction

2.1. 在 HS 编码预测方面的先前工作

Previous works treat HSCode prediction as the e-commerce text classification task [Grainger, 2024], using pre-trained BERT models [Liao et al., 2024, Shubham et al., 2022] or Large Language Models (LLMs) prompting [Hussain and Ahmed, 2023]. However, these approaches fail to leverage domain-specific knowledge, especially the rules written by human experts [Hussain and Ahmed, 2023, Judy, 2024]. Besides, existing HSCode benchmarks face two critical limitations [Judy, 2024, Lee et al., 2024, Stassin et al., 2023]: (1) they are typically constructed from publicly accessible customs rulings, suffering from data leakage; (2) they are not released. In contrast, our released HSCODECOMP is collected from large-scale online shopping platforms with noisy product descriptions, making it more challenging and realistic.

先前的研究将 HSCode 预测视为电子商务文本分类任务 [Grainger, 2024]，使用预训练的 BERT 模型 [Liao 等，2024, Shubham 等，2022] 或大型语言模型 (LLMs) 提示 [Hussain 和 Ahmed, 2023]。然而，这些方法未能充分利用领域专业知识，特别是由人类专家编写的规则 [Hussain 和 Ahmed, 2023, Judy, 2024]。此外，现有的 HSCode 基准数据集面临两个关键限制 [Judy, 2024, Lee 等，2024, Stassin 等，2023]：

(1) 它们通常基于公开的海关裁定构建，存在数据泄露问题；(2) 它们未被发布。相比之下，我们发布的 HSCODECOMP 是从大规模在线购物平台收集的，包含嘈杂的产品描述，使其更具挑战性和现实性。

2.2. Benchmarking Level 1 Knowledge Utilization

2.2. 基准测试级别 1 知识利用

Numerous benchmarks have been proposed to evaluate agent capabilities in understanding and deep reasoning over long-form open-domain web content [Thomas et al., 2025, Yao et al., 2024, Joshi et al., 2017, Phan et al., 2025]. For example, WebArena [Zhou et al., 2023] provides realistic, self-hostable websites with standardized evaluation protocols to assess functional correctness. WebShop [Yao et al., 2022] and ALFWorld [Shridhar et al., 2021] evaluate long-horizon decision-making abilities of agents in web environments through tool interactions. More recent deep search benchmarks, such as GAIA [Mialon et al., 2023a], BrowseComp [Wei et al., 2025], WebWalkerQA [Wu et al., 2025] and BrowseComp-ZH [Zhou et al., 2025], demand advanced tool-usage and deep reasoning capabilities [Zhang et al., 2025, Li et al., 2025b].

已提出了许多基准测试来评估代理在理解和对长文本开放领域网络内容进行深度推理的能力 [Thomas 等人, 2025, Yao 等人, 2024, Joshi 等人, 2017, Phan 等人, 2025]。例如，WebArena [Zhou 等人, 2023] 提供了现实的、可自行托管的网站，并采用标准化的评估协议来评估功能正确性。WebShop [Yao 等人, 2022] 和 ALFWorld [Shridhar 等人, 2021] 通过工具交互评估代理在网页环境中的长期决策能力。更近期的深度搜索基准测试，例如 GAIA [Mialon 等人, 2023a]、BrowseComp [Wei 等人, 2025]、WebWalkerQA [Wu 等人, 2025] 和 BrowseComp-ZH [Zhou 等人, 2025]，要求代理具备先进的工具使用和深度推理能力 [Zhang 等人, 2025, Li 等人, 2025b]。

2.3. Benchmarking Level 2 Knowledge Utilization

2.3. 二级知识利用基准测试

Recent works have focused on how agents utilize structured knowledge in domain-specific applications. Unlike open-domain data, domain-specific knowledge is typically organized into structured formats such as databases and knowledge graphs [Huang et al., 2025, Yu et al., 2025a, Chen et al., 2025a], enabling more precise knowledge retrieval and utilization. To evaluate these capabilities, numerous deep search benchmarks have been proposed, including WebMall [Peeters et al., 2025b] and DeepShop [Lyu et al., 2025] for e-commerce, LegalAgentBench for law [Li et al., 2025a], FinSearchComp for finance [Hu et al., 2025a], DAgent for data analysis [Xu et al., 2025], CRMarena for CRM workflows [Huang et al., 2025], and MedBrowseComp for medicine [Chen et al., 2025b].

近期的研究主要关注代理如何在特定领域应用中利用结构化知识。与开放领域数据不同，领域特定知识通常被组织为数据库和知识图谱等结构化格式 [Huang 等, 2025, Yu 等, 2025a, Chen 等, 2025a]，从而实现更精确的知识检索和利用。为了评估这些能力，已经提出了许多深度搜索基准测试，包括用于电子商务的 WebMall [Peeters 等, 2025b] 和 DeepShop [Lyu 等, 2025]，用于法律的 LegalAgentBench [Li 等, 2025a]，用于金融的 FinSearchComp [Hu 等, 2025a]，用于数据分析的 DAgent [Xu 等, 2025]，用于 CRM 工作流的 CRMarena [Huang 等, 2025]，以及用于医学的 MedBrowseComp [Chen 等, 2025b]。

In summary, while there exists numerous evaluation benchmarks for assessing agent performance in open-domain or domain-specific scenarios, none evaluates the ability to apply Level 3 abstract rule-based knowledge. To address this critical gap, we introduce a realistic and expert-level e-commerce benchmark HSCODECOMP. Our benchmark presents significant challenges even for state-of-the-art closed-source and open-source agent systems.

总之，尽管存在许多用于评估代理在开放领域或特定领域场景中性能的评估基准，但没有一个基准能够评估应用三级抽象规则知识的能力。为了解决这一关键差距，我们引入了一个现实且专家级的电子商务基准 HSCODECOMP。我们的基准即使对最先进的闭源和开源代理系统也提出了重大挑战。

3. Task Formulation of HSCode Prediction

3. HS 编码预测任务建模

The HSCode prediction task is to assign a valid and unique 10-digit Harmonized System (HS) code to a given noisy but realistic product description. The product HSCode plays the crucial role in e-commerce system. It is the global standard for classifying traded goods, essential for tariffs, customs, and trade governance. The core challenge is to learn a mapping function, i.e., agents implemented by Large Language Models (LLMs) or Vision Language Models (VLMs), $f: \mathcal{X} \rightarrow \mathcal{Y}$.

HS 编码预测任务是为给定的嘈杂但真实的商品描述分配一个有效且唯一的 10 位协调制度 (HS) 编码。商品的 HS 编码在电子商务系统中起着至关重要的作用。它是对 traded goods 进行分类的全球标准，对于关税、海关和贸易管理至关重要。核心挑战是学习一个映射函数，即由大型语言模型 (LLMs) 或视觉语言模型 (VLMs) 实现的代理， $f: \mathcal{X} \rightarrow \mathcal{Y}$ 。

Product title t
Green R36S Retro Handheld Video Game Console Linux System 3.5Inch IPS Screen R35s Pro Portable Pocket Video Player 64G 128G 256G

Price p 831.55
 Currency u CNY

Consumer Electronics
 Games & Accessories
 Handheld Game Players

Level1 category
 Level2 category
 Level3 category

Product categories c

Product URL r
<https://www.aliexpress.com/item/1005008091741434.html>

Product Attributes A

Attribute	Value	Attribute	Value	Attribute	Value	Attribute	Value
Origin	Mainland China	Bluetooth-compatible	No	Games Type	Retro Games	Screen Type	IPS
Screen Resolution	640x480	Games included	15000+	Display Size	3.5"	WIFI	No
Category	Handheld Game Players	High-concerned Chemical	None	Type of devices	other	Sold in	sell_by_piece
Measurement unit	100000015	Brand Name	Bokiddy	Operating System	Linux	Supporting Language	Brazilian Portuguese, ...
RGB lighting joysticks	No	each pack	1	Package	Yes	Battery Capacity[mAh]	3000

Product Image i

HSCode y
9504904000
Chapter
 ↳ Heading
 ↳ Sub-Heading
 ↳ Country-specific

Figure 3: images/e10e45cf7bdf57cf5ffa06a448b0b83c81d1037efc6cf22faeaa7bb1100bcc3e.jpg

Figure 2. One example of a game console product in HSCODECOMP.

图 2. HSCODECOMP 中的一个游戏机产品示例。

Input: Figure 2 shows that each product $x \in \mathcal{X}$ encompass rich records: $x = (t, A, c, i, p, u, r)$, where t is the product title; $A = \{(k_j, v_j)\}_{j=1}^K$ represents a set of K product attributes (e.g., material and package size); c represents product categories defined by the e-commerce platform; i is the product image; p and u are the price and currency; and r is the webpage URL of the product.

图 2 显示，每个产品 $x \in \mathcal{X}$ 包含丰富的记录： $x = (t, A, c, i, p, u, r)$ ，其中 t 是产品标题； $A = \{(k_j, v_j)\}_{j=1}^K$ 表示一组 K 产品属性（例如材料和包装尺寸）； c 表示由电商平台定义的产品类别； i 是产品图片； p 和 u 是价格和货币； r 是产品的网页 URL。

Hierarchical Rule Utilization: Accurate HSCode prediction requires effectively utilizing three

types of e-commerce knowledge: (1) Hierarchical tariff rules from official classification systems (e.g., eWTP), which organize product categories in a hierarchical structure. As shown in Figure 11, these rules contain complex implicit logic relationships, for example, the exception clause in tariff rules like excluding articles of HS heading 8539 ... (highlighted with red boxes). Besides, these rules often employ vague linguistic constraints (highlighted with blue boxes) that challenge existing AI agents; (2) Human-written decision rules that specify how to correctly apply tariff rules. These rules provide high-level decision principles (see Figure F.1 for an example with six key principles defined by domain experts); and (3) Official customs rulings databases, such as the U.S. Customs Rulings Online Search System (CROSS) ², which document historical HSCode classification decisions. As illustrated in Figure 12, these databases contain complex information format requiring advanced reasoning capabilities.

分层规则利用：准确的 HS 编码预测需要有效利用三种类型的电子商务知识：(1) 来自官方分类系统的分层关税规则（例如 eWTP），这些规则以分层结构组织产品类别。如图 11 所示，这些规则包含复杂的隐含逻辑关系，例如关税规则中的例外条款，例如排除 HS 编码 8539 项下的商品.....（用红色方框突出显示）。此外，这些规则经常使用模糊的语义约束（用蓝色方框突出显示），这给现有的 AI 代理带来了挑战；(2) 人工编写的决策规则，这些规则指明如何正确应用关税规则。这些规则提供了高层次的决策原则（例如图 F.1 中由领域专家定义的六个关键原则）；(3) 官方海关裁定数据库，例如美国海关裁定在线搜索系统（CROSS）²，这些数据库记录了历史 HS 编码分类决策。如图 12 所示，这些数据库包含需要高级推理能力的复杂信息格式。

Output: The HSCode $y \in Y$ is a single 10-digit numeric string $Y \subseteq \{0, 1, \dots, 9\}^{10}$. The HSCode is hierarchical, where first 2-digit, 4-digit, and 6-digit represents the HS chapter, heading and sub-heading of tariff classification of products, respectively, and last 4 digits (from 6 to 10) are country-specific codes. In summary, this 10-digit HSCode follows a valid path in the official HS taxonomy.

输出：HS 编码 $y \in Y$ 是一个 10 位数字字符串 $Y \subseteq \{0, 1, \dots, 9\}^{10}$ 。HS 编码具有层次结构，其中前两位、前四位和前六位分别代表税则归类商品的章、目和子目，而最后四位（从第 6 位到第 10 位）是国别特定代码。总之，这个 10 位 HS 编码遵循官方 HS 分类法中的有效路径。

4. HSCODECOMP Construction and Evaluation Metrics

4. HSCODECOMP 构建与评估指标

4.1. Benchmark Construction

4.1. 基准构建

We design a rigorous pipeline, ensuring the dataset is diverse, realistic and expert-level: (1) Data Collection and Diversity Control; (2) Human Expert Annotation; and (3) Human Expert Validation.

我们设计了一个严谨的流程，确保数据集具有多样性、真实性和专家级水平：(1) 数据收集与多样性控制；(2) 人工专家标注；以及 (3) 人工专家验证。

Data Collection and Diversity Control. Products in our proposed HSCODECOMP is sourced from a large-scale global e-commerce platform. These product profiles include the noisy information, ensuring that task instances reflect the real-world challenges. Besides, we also balance the data category distribution to prevent the potential topical skew. Specifically, we apply a pre-processing step: a semantic redundancy filter discards the products (x) sharing identical categories (c) and 10-digit HSCode (y) with existing product instances. This ensures that HSCODECOMP is not dominated by common and easy-to-classify products.

数据收集与多样性控制。我们提出的 HSCODECOMP 中的产品来源于一个大规模的全球电商平台。这些产品资料包含噪声信息，确保任务实例能够反映现实世界中的挑战。此外，我们还平衡了数据类别分布，以防止潜在的主题偏差。具体来说，我们应用了一个预处理步骤：语义冗余过滤器会丢弃与现有产品实例

具有相同类别 (c) 和 10 位 HS 编码 (y) 的产品 (x)。这确保了 HSCODECOMP 不会被常见且容易分类的产品所主导。

Human Expert Annotation. To ensure the quality of HSCODECOMP, we engage human experts specialized in HSCode classification to annotate the HSCode (y) for each product profile (x). As shown in Figure 3 (left), the annotation process follows a five-step pipeline: (1) Two experts gather comprehensive information from the product webpage (Step 1); (2) They extract the core structured features of products (Step 2); (3) Experts search the official customs ruling databases (CROSS) for related cases. If a related case is found (very rare during the annotation of HSCODECOMP), the corresponding HSCode is then validated on eWTP system, followed by the minor revision. Otherwise, they refine their search queries or revisit Step 2 to adjust the extracted features (Step 3); (4) For products without any related cases, experts execute human-written hierarchical decision rules to apply tariff rules, and determine the appropriate HSCode (Step 4); (5) Finally, experts verify the final identified HSCodes on the eWTP website to ensure its validity (Step 5).

人工专家标注。为确保 HSCODECOMP 的质量，我们聘请了专门从事 HS 编码分类的人工专家，对每个产品档案 (x) 的 HS 编码 (y) 进行标注。如图 3（左）所示，标注过程遵循一个五步流程：（1）两位专家从产品网页中收集全面的信息（步骤 1）；（2）他们提取产品的核心结构化特征（步骤 2）；（3）专家在官方海关裁定数据库（CROSS）中搜索相关案例。如果找到相关案例（在 HSCODECOMP 的标注过程中非常罕见），则在 eWTP 系统中验证相应的 HS 编码，随后进行小幅修订。否则，他们将调整搜索查询或返回步骤 2 以调整提取的特征（步骤 3）；（4）对于没有任何相关案例的产品，专家执行人工编写的分层决策规则，应用关税规则并确定适当的 HS 编码（步骤 4）；（5）最后，专家在 eWTP 网站上验证最终确定的 HS 编码，以确保其有效性（步骤 5）。

Human Expert Validation. As shown in the Figure 3 (Step 6), when the HSCodes assigned by two experts match, the code is accepted. When they disagree, a senior tariff expert reviews both annotations to determine the correct HSCode. If neither annotation is valid, the instance is excluded from the dataset. Finally, we collect 632 products with their human-annotated corresponding HSCodes, spanning 32 first-level product category defined by a large-scale ecommerce platform and 27 HS chapters defined by eWTP³. Furthermore, to verify the reliability of our process, we conducted an additional quality review. A fourth senior expert, not involved in the initial annotation, re-annotated a random 10% sample of the dataset. This review shows only a 2% disagreement rate, confirming the effectiveness and consistency of our dataset construction pipeline.

人工专家验证。如图 3（步骤 6）所示，当两位专家分配的 HS 编码一致时，该编码将被接受。当他们意见不一致时，一位高级关税专家将审查两个注释，以确定正确的 HS 编码。如果两个注释均无效，则该实例将被排除在数据集之外。最后，我们收集了 632 个产品及其对应的人工注释 HS 编码，涵盖了一个大型电商平台定义的 32 个一级产品类别和 eWTP 定义的 27 个 HS 章节³。此外，为了验证我们流程的可靠性，我们进行了额外的质量审查。一位未参与初始注释的第四位高级专家重新标注了数据集的随机 10% 样本。此次审查显示仅存在 2% 的分歧率，证实了我们数据集构建流程的有效性和一致性。

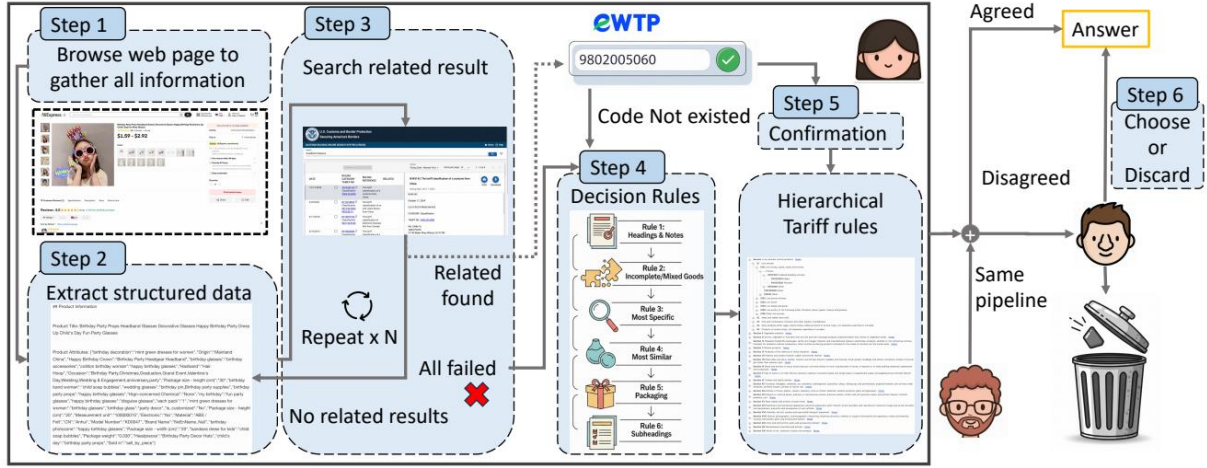


Figure 4: images/34635163fea650341c024d47af26147800e9cdc6f04f984fc9b33df6d038238b5.jpg

Figure 3. The pipeline for human experts to annotate the HS Codes, including two human experts for HS Code annotation (Step 1 to 5) and one additional expert for quality validation (Step 6).

图 3. 人类专家标注 HS Codes 的流程，包括两位负责 HS Code 标注的人类专家（步骤 1 至 5）和一位负责质量验证的额外专家（步骤 6）。

4.2. Evaluation Metric

4.2. 评估指标

We conduct the exact match to compare the normalized HS Codes extracted from the final output against human-annotated ground truth. Our primary evaluation metric is the 10-digit HS Code accuracy, which measures whether the predicted code exactly matches the reference 10-digit code. Additionally, we also report accuracies at 2-digit, 4-digit, 6-digit and 8-digit levels to provide more comprehensive insights into the performance across different granularities.

我们进行精确匹配，以将从最终输出中提取的标准化 HS Codes 与人工标注的真实值进行比较。我们的主要评估指标是 10 位 HS 编码的准确性，用于衡量预测的编码是否与参考的 10 位编码完全一致。此外，我们还报告了 2 位、4 位、6 位和 8 位级别的准确性，以提供更全面的性能洞察，涵盖不同粒度的水平。

5. Experiments

5. 实验

5.1. Experimental Setup

5.1. 实验设置

We evaluate three kinds of advanced approaches on HSCODECOMP:

我们在 HSCODECOMP 上评估了三种先进的方法：

LLMs/VLMs (no tools): We test 14 foundation models (GPT-5, Gemini-2.5-PRO, GPT-4o, Kimi-K2 [Team et al., 2025], Claude Sonnet 4, DeepSeek series [DeepSeek-AI et al., 2025], Qwen variants

[Yang et al., 2025], O3-mini, Nemotron-32B [Bercovich et al., 2025]) for HSCode prediction using only internal knowledge. For VLMs (GPT-4o, GPT-5, Claude Sonnet 4, Gemini 2.5 Pro), we provide product images to assess the impact of the visual information.

LLMs/VLMs (无工具): 我们使用 14 个基础模型 (GPT-5、Gemini-2.5-PRO、GPT-4o、Kimi-K2 [Team 等, 2025]、Claude Sonnet 4、DeepSeek 系列 [DeepSeek-AI 等, 2025]、Qwen 变体 [Yang 等, 2025]、O3-mini、Nemotron-32B [Bercovich 等, 2025]) 进行 HSCode 预测, 仅使用内部知识。对于 VLMs (GPT-4o、GPT-5、Claude Sonnet 4、Gemini 2.5 Pro), 我们提供产品图片以评估视觉信息的影响。

Open-source Agent Systems: We evaluate six open-source frameworks (SmolAgents [Roucher et al., 2025], Aworld [Yu et al., 2025c], Agentorchestra [Zhang et al., 2025], OWL [Hu et al., 2025b], WebSailor [Li et al., 2025b] and Cognitive Kernel [Fang et al., 2025]) using GPT-5 as the default backbone. SmolAgent is enhanced with vision capabilities via product images, while Vision Language Models are incompatible with other agent frameworks. All frameworks utilize standardized tools including web search.

开源代理系统: 我们使用 GPT-5 作为默认骨干模型, 评估了六个开源框架 (SmolAgents [Roucher 等, 2025], Aworld [Yu 等, 2025c], Agentorchestra [Zhang 等, 2025], OWL [Hu 等, 2025b], WebSailor [Li 等, 2025b] 和 Cognitive Kernel [Fang 等, 2025])。SmolAgent 通过产品图片增强了视觉能力, 而视觉语言模型与其他代理框架不兼容。所有框架都使用标准化工具, 包括网络搜索。

Closed-source Agent Systems: We assess the performance of commercial systems Manus, Gemini Deep Research, and Grok DeepSearch. As these systems do not provide public APIs, we conduct

闭源代理系统: 我们评估了商业系统 Manus、Gemini Deep Research 和 Grok DeepSearch 的性能。由于这些系统未提供公开 API, 我们进行

Baselines	Model Type	HSCode Prediction Accuracy			
2-digit	4-digit	6-digit	8-digit	10-digit	
LLM/VLM-Only					
GPT-5	VLM	82.12	70.89	59.97	
Gemini-2.5-PRO	VLM	82.28	71.04	59.02	
GPT-4o	VLM	78.01	64.08	48.10	
Claude Sonnet 4	VLM	78.80	64.08	45.25	
GPT-5	LLM	82.59	69.78	56.33	
Gemini-2.5-PRO	LLM	80.54	69.94	58.54	
GPT-4o	LLM	75.47	61.55	45.73	
Claude Sonnet 4	LLM	78.80	62.97	44.94	
Kimi-K2	LLM	78.01	62.03	44.15	
DeepSeek-R1	LLM	77.22	61.71	38.45	
DeepSeek-V3	LLM	77.06	54.43	32.28	
Qwen-MAX	LLM	71.52	48.58	24.21	
Qwen3-235B-A22B	LLM	66.93	49.53	24.53	
O3-mini	LLM	77.22	56.17	24.53	
Qwen3-32B	LLM	64.40	29.27	8.07	
QWQ-32B	LLM	66.77	29.11	4.43	
Qwen2.5-72B	LLM	20.73	12.34	3.80	
Nemotron-32B	LLM	43.51	5.70	0.16	
Open-source Agent System (GPT-5 Backbone)					

SmolAgents	VLM	82.06	72.06	62.38
SmolAgents	LLM	82.28	70.89	59.81
Aworld	LLM	82.28	70.41	59.18
Agentorchestra	LLM	82.12	70.73	60.44
OWL	LLM	72.63	61.87	51.58
WebSailor	LLM	81.64	70.56	57.27
Cognitive Kernel	LLM	80.06	69.15	54.59

基线	模型类型	HS 编码预测准确率				
2 位数字	4 位数字	6 位数字	8 位数字	10 位数字		
LLM/VLM-Only						
GPT-5	VLM	82.12	70.89	59.97	41.46	29.27
Gemini-2.5-PRO	VLM	82.28	71.04	59.02	40.51	24.21
GPT-4o	VLM	78.01	64.08	48.10	29.75	18.51
十四行诗 4	VLM	78.80	64.08	45.25	22.63	11.23
GPT-5	大型语言模型	82.59	69.78	56.33	40.98	28.96
Gemini-2.5-PRO	大型语言模型	80.54	69.94	58.54	40.35	23.42
GPT-4o	大型语言模型	75.47	61.55	45.73	30.06	18.35
莎士比亚十四行诗第四首	大语言模型	78.80	62.97	44.94	23.58	11.87
Kimi-K2	大型语言模型	78.01	62.03	44.15	24.53	12.18
DeepSeek-R1	大型语言模型	77.22	61.71	38.45	16.77	6.65
DeepSeek-V3	大型语言模型	77.06	54.43	32.28	17.25	6.49
Qwen-MAX	大语言模型	71.52	48.58	24.21	11.23	3.80
Qwen3-235B-A22B	大型语言模型	66.93	49.53	24.53	6.01	1.74
O3-mini	大型语言模型	77.22	56.17	24.53	6.65	1.27
Qwen3-32B	大语言模型	64.40	29.27	8.07	1.27	0.32
QWQ-32B	大型语言模型	66.77	29.11	4.43	1.42	0.16
Qwen2.5-72B	大型语言模型	20.73	12.34	3.80	1.42	0.16
Nemotron-32B	大型语言模型	43.51	5.70	0.16	0.00	0.00
开源代理系统 (GPT-5 主干)						
SmolAgents	VLM	82.06	72.06	62.38	52.38	46.83
SmolAgents	大型语言模型	82.28	70.89	59.81	49.05	42.72
一个世界	大语言模型	82.28	70.41	59.18	48.58	41.30
Agentorchestra	大型语言模型	82.12	70.73	60.44	47.78	41.30
OWL	大型语言模型	72.63	61.87	51.58	41.77	37.34
WebSailor	大型语言模型	81.64	70.56	57.27	43.98	35.44
认知内核	大型语言模型	80.06	69.15	54.59	40.03	26.42

Table 1. The complete results of state-of-the-art baselines in our proposed HSCODECOMP.

表 1. 我们提出的 HSCODECOMP 中最新基线的完整结果。

manual evaluations on 49 representative examples from the HSCODECOMP benchmark, following the evaluation protocol established in prior work [Li et al., 2025b].

按照之前工作的评估协议 [Li 等人, 2025b], 在 HSCODECOMP 基准的 49 个代表性示例上进行人工评估。

All systems produce standardized outputs: a single HSCode in `[...]` format. More implementation details appear in Appendix F.

所有系统都会产生标准化的输出：以 `[...]` 格式呈现的一个 HSCode。更多实施细节请参见附录 F。

5.2. Main Results

5.2. 主要结果

Table 1 summarizes the performance of state-of-the-art LLM/VLM-Only models and open-source agents on HSCODECOMP. All approaches exhibit a consistent decline in accuracy as the HSCode length increases. Notably, LLM/VLM-only baselines are much worse than agent systems due to their lack of domain-specific knowledge. The best baseline, SmolAgent (GPT-5 VLM version), achieves only 46.83% 10-digit accuracy, which remains substantially below the 95% accuracy achieved by experienced human experts. To ensure a fair comparison, we eval-

表 1 总结了当前最先进的 LLM/VLM-Only 模型和开源代理在 HSCODECOMP 上的性能。所有方法在 HSCode 长度增加时都表现出准确率的持续下降。值得注意的是，由于缺乏领域特定知识，仅使用 LLM/VLM 的基线方法表现明显不如代理系统。最好的基线模型 SmolAgent（GPT-5 VLM 版本）仅实现了 46.83% 的 10 位数字准确率，这仍然显著低于经验丰富的专家达到的 95% 准确率。为了确保公平比较，我们评估了...

Agent Systems	10-digit
Gemini Deep Researcher	40.81
Manus	30.61
Grok DeepSearch	26.53
SmolAgents (GPT-5)	42.86
Aworld (GPT-5)	42.86

代理系统	10 位数字
Gemini Deep Researcher	40.81
手稿	30.61
Grok DeepSearch	26.53
SmolAgents (GPT-5)	42.86
Aworld (GPT-5)	42.86

Table 2. Comparison between closed-source and open-source agents.

表 2. 闭源代理与开源代理的比较。

uate both closed-source and open-source agents on the same subset of HSCODECOMP. Table 2 shows open-source agents outperform closed-source agents. Case studies reveal that closed-source agents suffer from the premature decisions and information misprocessing problems, as detailed in Section 6.3. In summary, the significant performance gap between human experts

and the top-performing agent system underscores the challenges presented by HSCODECOMP. To better understand the factors affecting the performance, we conduct three ablation studies as below.

在相同的 HSCODECOMP 子集上评估了闭源和开源代理。表 2 显示开源代理的表现优于闭源代理。案例研究揭示了闭源代理存在过早决策和信息误处理的问题，如第 6.3 节所详细说明的。总之，人类专家与表现最好的代理系统之间显著的性能差距突显了 HSCODECOMP 所带来的挑战。为了更好地理解影响性能的因素，我们进行了以下三项消融研究。

Ablation Study on Hierarchical Decision Rules The hierarchical decision rules capture how human experts apply tariff rules. To assess whether agents can effectively leverage these rules, we conduct ablation experiments for following models: GPT-5, SmolAgent (GPT-5 VLM version), Aword (GPT-5 LLM version) and WebSailor (GPT-5 LLM version). As shown in Table 3, incorporating decision rules (w/ DR) decreases accuracy for both SmolAgent and WebSailor, while Aword achieves only marginal

分层决策规则的消融研究 分层决策规则捕捉了人类专家如何应用关税规则。为了评估智能体是否能够有效利用这些规则，我们对以下模型进行了消融实验：GPT-5、SmolAgent (GPT-5 VLM 版本)、Aword (GPT-5 LLM 版本) 和 WebSailor (GPT-5 LLM 版本)。如表 3 所示，引入决策规则（带 DR）会使 SmolAgent 和 WebSailor 的准确性下降，而 Aword 仅实现了微小的提升。

gains. Therefore, we remove these decision rules for agents as the default setup. These results indicate that current agent systems struggle at applying human-written decision rules, thereby limits their ability to utilize hierarchical tariff rules for HSCode prediction.

收益。因此，我们将这些决策规则从代理的默认设置中移除。这些结果表明，当前的代理系统在应用人工编写的决策规则方面存在困难，从而限制了它们利用分层关税规则进行 HSCode 预测的能力。

Backbone Model	Model Type	10-digit
SmolAgent w/o DR	VLM	46.83
SmolAgent w/ DR	VLM	43.83 □
Aworld w/o DR	LLM	41.30
Aworld w/ DR	LLM	42.95 □
WebSailor w/o DR	LLM	35.44
WebSailor w/ DR	LLM	35.43 □

主干模型	模型类型	10 位数字
SmolAgent w/o DR	VLM	46.83
SmolAgent w/ DR	视觉语言模型 (VLM)	43.83 □
一个没有 DR 的世界	大语言模型	41.30
一个有 DR 的世界	大语言模型	42.95 □
WebSailor 无 DR	大型语言模型	35.44
WebSailor w/ DR	大型语言模型	35.43 □

Table 3. The ablation study on human-written Decision Rules (DR). GPT-5 is the backbone.

表 3. 对人工编写的决策规则（DR）的消融研究。GPT-5 是主干模型。

Multi-modal Information Is Helpful Table 1 and Table 4 show that most baselines achieve consistent improvements when the product images can be accessed. Case studies in Appendix I show

that understanding product images improves performance by capturing visual attributes—such as material and surface features—that are not present in the textual description but are critical for classification. These attributes align with the predefined rules, leading to performance gains.

多模态信息是有帮助的表 1 和表 4 显示，当可以访问产品图像时，大多数基线模型都能实现一致的性能提升。附录 I 中的案例研究显示，理解产品图像通过捕捉文本描述中没有但对分类至关重要的视觉属性（如材质和表面特征）来提高性能。这些属性与预定义的规则相一致，从而带来性能的提升。

Backbone Model	10-digit
GPT-5 w/o Image	42.72
GPT-5 w/ Image	46.83 $\square$
Gemini-2.5-Pro w/o Image	34.49
Gemini-2.5-Pro w/ Image	34.39 $\square$
Claude 4 Sonnet w/o Image	33.70
Claude 4 Sonnet w/ Image	34.65 $\square$
GPT-4o w/o Image	22.03
GPT-4o w/ Image	22.31 $\square$

主干模型	10 位数字
GPT-5 无图像	42.72
GPT-5 w/ Image	46.83 $\square$
Gemini-2.5-Pro 无图像版本	34.49
Gemini-2.5-Pro w/ Image	34.39 $\square$
Claude 4 Sonnet 无图像	33.70
Claude 4 Sonnet w/ Image	34.65 $\square$
GPT-4o 无图像	22.03
GPT-4o w/ Image	22.31 $\square$

Webpage Visits Decrease Agents Performance We augment SmolAgent (GPT-5 LLM version) with the capability to visit webpages, but this leading to 10-digit accuracy

网页访问降低代理性能我们为 SmolAgent（GPT-5 大型语言模型版本）增加了访问网页的能力，但这导致了 10 位数的精度下降

decrease from 42.72% to 42.09%. Our study reveals that a large amount of webpage content overwhelms the key information, misleading the agents, while this key information can be precisely extract by search engines in the snippets. Therefore, we remove the webpage visit tool for all open-source agents as the default setup.

下降至 42.09%。我们的研究显示，大量网页内容会掩盖关键信息，误导代理，而这些关键信息可以通过搜索引擎在摘要中精确提取。因此，我们默认移除所有开源代理的网页访问工具。

Table 4. The ablation study of the product images in SmolAgents.

表 4. SmolAgents 中产品图像的消融研究。

6. Analysis on HSCodeCOMP

6. 对 HSCodeCOMP 的分析

We conduct several detailed analysis: (1) Overthinking Decrease Open-Source Agent Performance (Section 6.1); (2) Effects of Backbones in agents (Section 6.2); (3) Failure Modes of Closed-Source and Open-Source Agents (Section 6.3); (4) Per-category performance analysis (Section 6.4); and (5) Effectiveness of Test-time Scaling (Section 6.5).

我们进行了几项详细分析：(1) 过度思考会降低开源代理的性能（第 6.1 节）；(2) 代理中主干网络的影响（第 6.2 节）；(3) 闭源和开源代理的失败模式（第 6.3 节）；(4) 按类别进行的性能分析（第 6.4 节）；以及 (5) 测试时扩展的有效性（第 6.5 节）。

6.1. Overthinking Decrease Open-source Agent Performance

6.1. 过度思考会降低开源代理的性能

Table 1 shows WebSailor underperforms SmolAgent despite using identical models and tools. Our case studies reveal that this occurs because WebSailor encourages excessive reasoning (Overthink), before tool-calling. Cases in Appendix H show that WebSailor often first conduct deep reasoning to predict the full 10-digit

表 1 显示，尽管 WebSailor 使用了与 SmolAgent 相同的模型和工具，但其表现仍不如 SmolAgent。我们的案例研究揭示了这一现象的原因：WebSailor 在调用工具之前会鼓励进行过多的推理（Overthink）。附录 H 中的案例表明，WebSailor 经常首先进行深入推理以预测完整的 10 位数字。

Agent Systems	Think Depth	10-digit
SmolAgent (GPT-5)	No-Think	42.72
WebSailor (GPT-5)	No-Think	40.82
WebSailor (GPT-5)	Medium-Think	37.34
WebSailor (GPT-5)	Overthink	35.44

代理系统	深度思考	10 位数字
SmolAgent (GPT-5)	不思考	42.72
WebSailor (GPT-5)	不思考	40.82
WebSailor (GPT-5)	中等思考	37.34
WebSailor (GPT-5)	过度思考	35.44

Table 5. Agent performance with different think depth. GPT-5 LLM version is the backbone. HSCode. The errors in reasoning significantly decrease the effectiveness of tool-calling. To prove this, we created two variants for WebSailor: (1) No-Think: direct tool calling without thinking; and (2) Medium-Think: medium-level reasoning depth before tool-calling. Medium-think denotes a moderate reasoning depth between No-think and Overthink. Table 5 demonstrates that reducing reasoning depth improves accuracy, with No-Think nearly matching SmolAgent. This finding suggests that the primary factor contributing to performance variances among open-source agents is the reasoning depth defined in the task prompt. For HSCode prediction, minimal reasoning with frequent tool calls outperforms extensive self-reasoning. When accurate information is available through calling tools, prioritizing tool utilization over reasoning yields better results for such complex domain-specific tasks.

表 5. 不同思考深度下的代理性能。GPT-5 LLM 版本是基础模型。

HSCode。推理中的错误显著降低了工具调用的有效性。为了证明这一点，我们为 WebSailor 创建了两个变体：(1) No-Think：直接调用工具而不进行思考；(2) Medium-Think：在调用工具前进行中等层次的推理。Medium-Think 表示介于 No-Think 和 Overthink 之间的适度推理深度。表 5 显示，减少推理深度可以提高准确性，其中 No-Think 几乎与 SmolAgent 相当。这一发现表明，开源代理之间性能差异的主要因素是任务提示中定义的推理深度。对于 HSCode 预测，频繁调用工具的最小推理优于广泛的自我推理。当通过调用工具获取准确信息时，优先使用工具而非进行推理，对于此类复杂的领域特定任务会产生更好的结果。

6.2. Effect of Backbones in Open-source Agent Systems

6.2. 在开源代理系统中主干网络的影响

This subsection analyze the effects of the backbone LLM in the performance of agent systems. Specifically, we evaluate the performance of SmolAgents system implemented by four different backbone LLMs: GPT-5, Gemini-2.5-pro, Claude-4-Sonnet and Qwen-MAX. As demonstrated in Table 6, different backbone LLMs yield markedly different results in the HSCode prediction task, and GPT-5 is the best backbone model. Therefore, we choose GPT-5 as the default setup for open-source agents. More results are provided in Appendix D.

本小节分析了骨干大语言模型（LLM）对代理系统性能的影响。具体而言，我们评估了由四种不同的骨干 LLM 实现的 SmolAgents 系统的性能：GPT-5、Gemini-2.5-pro、Claude-4-Sonnet 和 Qwen-MAX。如表 6 所示，在 HSCode 预测任务中，不同的骨干 LLM 产生了显著不同的结果，其中 GPT-5 是表现最好的骨干模型。因此，我们选择 GPT-5 作为开源代理系统的默认配置。更多结果请参见附录 D。

Backbone Model	10-digit
GPT-5	42.72
Gemini-2.5-Pro	34.49
Claude 4 Sonnet	33.70
Qwen-MAX	17.43

主干模型	10 位数字
GPT-5	42.72
Gemini-2.5-Pro	34.49
Claude 4 Sonnet	33.70
Qwen-MAX	17.43

Table 6. The ablation study on the backbone models in SmolAgent.

表 6. SmolAgent 中主干模型的消融研究。

6.3. Failure Modes of Closed-Source and Open-Source Agents

6.3. 闭源代理和开源代理的失效模式

We perform the qualitative and quantitative analysis for closed-source and open-source agents.

我们对闭源和开源代理进行定性和定量分析。

Qualitative analysis. We identify six critical failure modes of open-source and closed-source agent systems in HSCODECOMP: (1) Premature Decisions: Agents commit to incorrect classification paths without collecting sufficient evidence (Table 9, Grok DeepSearch); (2) Information Misprocessing: Agents overlook or misinterpret key product details, indicating challenges with long-context processing (Table 8 and Figure 15); (3) Unnecessary Self-Correction: Agents sometimes predict correct HS Codes initially but revise them incorrectly through excessive critique (Table 8, Gemini Deep Research); (4) Reasoning Hallucination: Agents generate plausible but factually incorrect reasoning steps (Table 8, Grok DeepSearch); (5) Wrong Rule Application: Models frequently miss or misuse relevant tariff rules due to their ambiguous descriptions that confuse the reasoning process, resulting in incorrect classification decisions (Figure 17); and (6) Lack of Domain Knowledge: Models exhibit errors due to insufficient domain-specific knowledge, such as misidentify silicone products as rubber instead of plastic (Figure 16). These limitations highlight that HSCODECOMP remains challenging for advanced

定性分析。我们在 HSCODECOMP 中识别出开源和闭源代理系统存在六种关键的失败模式：(1) 过早决策：代理在未收集足够证据的情况下就承诺错误的分类路径（表 9, Grok DeepSearch）；(2) 信息误处理：代理忽略或误解关键产品细节，表明其在处理长上下文时存在困难（表 8 和图 15）；(3) 不必要的自我修正：代理有时最初正确预测了 HS Codes，但通过过度批评而错误地进行了修改（表 8, Gemini Deep Research）；(4) 推理幻觉：代理生成看似合理但事实错误的推理步骤（表 8, Grok DeepSearch）；(5) 错误应用规则：由于相关关税规则描述模糊，模型经常遗漏或误用这些规则，导致推理过程混乱，从而做出错误的分类决策（图 17）；以及 (6) 缺乏领域知识：由于缺乏特定领域的知识，模型会出现错误，例如将硅胶产品误认为橡胶而非塑料（图 16）。这些局限性表明，HSCODECOMP 对于高级模型来说仍然具有挑战性。

closed-source and open-source systems.

闭源系统和开源系统。

Quantitative analysis: Figure 4 present the distribution of four coarse-grained failures across both LLM/VLM-Only (left) and agents (right)⁴: (1) Outdated: Incorrect HS Codes due to changes in tariff rules over time; (2) Hallucination: Invalid HS Codes that do not exist in the official coding system; (3) Error but Valid: HS Codes are valid and current, but differ from the ground-truth HS Codes; and (4) Others: Other errors like wrong output formats, reaching maximum window and wrong tool-calling. Our analysis reveals that agents significantly reduce hallucination, outdated and other errors through effective tool utilization, compared with LLMs. Consequently, the predominant error type for agents is “Error but Valid”. Besides, Figure 9 also quantifies the improvements from GPT-5 to SmolAgent (GPT-5), demonstrating that the agent significantly reduces both outdated and hallucination errors.

定量分析：图 4 展示了四种粗粒度错误在 LLM/VLM-Only (左) 和代理 (右) 中的分布情况⁴：(1) 过时：由于关税规则随时间变化而导致的错误 HS Codes；(2) 幻觉：在官方编码系统中不存在的无效 HS Codes；(3) 错误但有效：HS Codes 有效且最新，但与真实 HS Codes 不同；以及 (4) 其他：如输出格式错误、达到最大窗口和错误调用工具等其他错误。我们的分析表明，与 LLM 相比，通过有效利用工具，代理显著减少了幻觉、过时和其他类型的错误。因此，代理的主要错误类型是“错误但有效”。此外，图 9 还量化了从 GPT-5 到 SmolAgent (GPT-5) 的改进情况，表明代理显著减少了过时和幻觉错误。

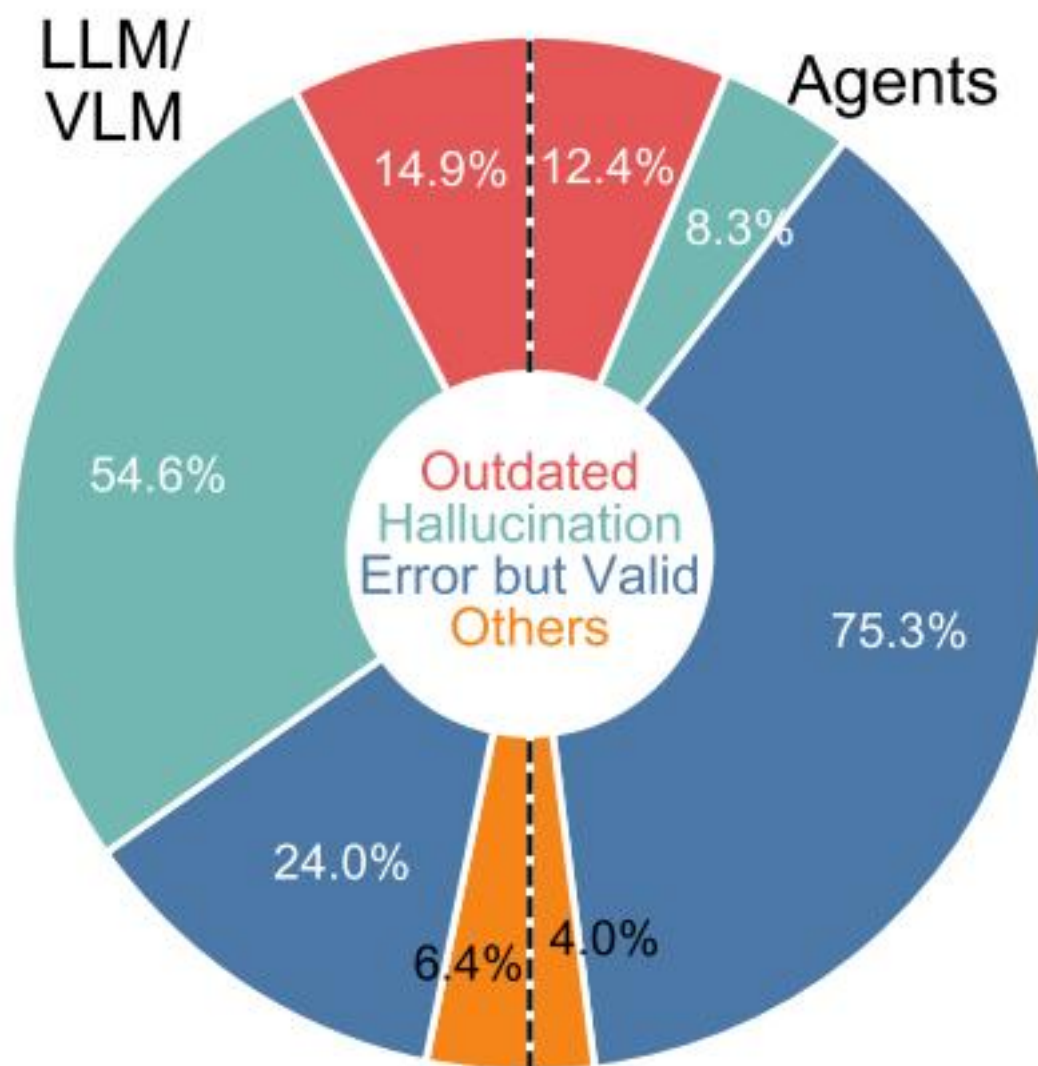


Figure 5: images/1c57b4c1df1d94f90bad9e8f4fb856e3e85ba2500d96d6d6d4fc94b20f5e9756.jpg

Figure 4. Failures analysis.

图 4. 失败分析。

6.4. Per-category Performance Analysis

6.4. 分类性能分析

We analyze two critical distributions across the 32 first-level product categories: (1) Challenging Product Distribution (CID): the distribution of products that all baseline methods failed to correctly predict; (2) Average Performance Distribution (APD): the distribution of average 10-digit accuracy across all baseline methods. Figure 5 reveals two key insights: (1) The CID indicates that the most challenging products are concentrated in long-tail categories, such as Novelty & Special Use (1.3%) and Men's Clothing (2.2%); (2) The APD shows that average accuracy across most product categories remains below 25%, with only Hair Extensions & Wigs achieving a relatively high accuracy of 47%. Importantly, even for most frequent categories like Jewelry & Accessories (13.1%), Home & Garden (16.8%) and Tools (6.2%), the

我们分析了 32 个一级产品类别中的两个关键分布：(1) 挑战性产品分布（CID）：所有基线方法均未能正确

预测的产品分布；(2) 平均性能分布 (APD)：所有基线方法的平均 10 位数字准确率分布。图 5 揭示了两个关键见解：(1) CID 表明最具挑战性的产品集中在长尾类别中，例如新颖性与特殊用途 (1.3%) 和男士服装 (2.2%)；(2) APD 显示，大多数产品类别的平均准确率仍低于 25%，只有假发与发片达到了相对较高的准确率 47%。重要的是，即使对于像珠宝与配饰 (13.1%)、家居与园艺 (16.8%) 和工具 (6.2%) 这样的高频类别，其

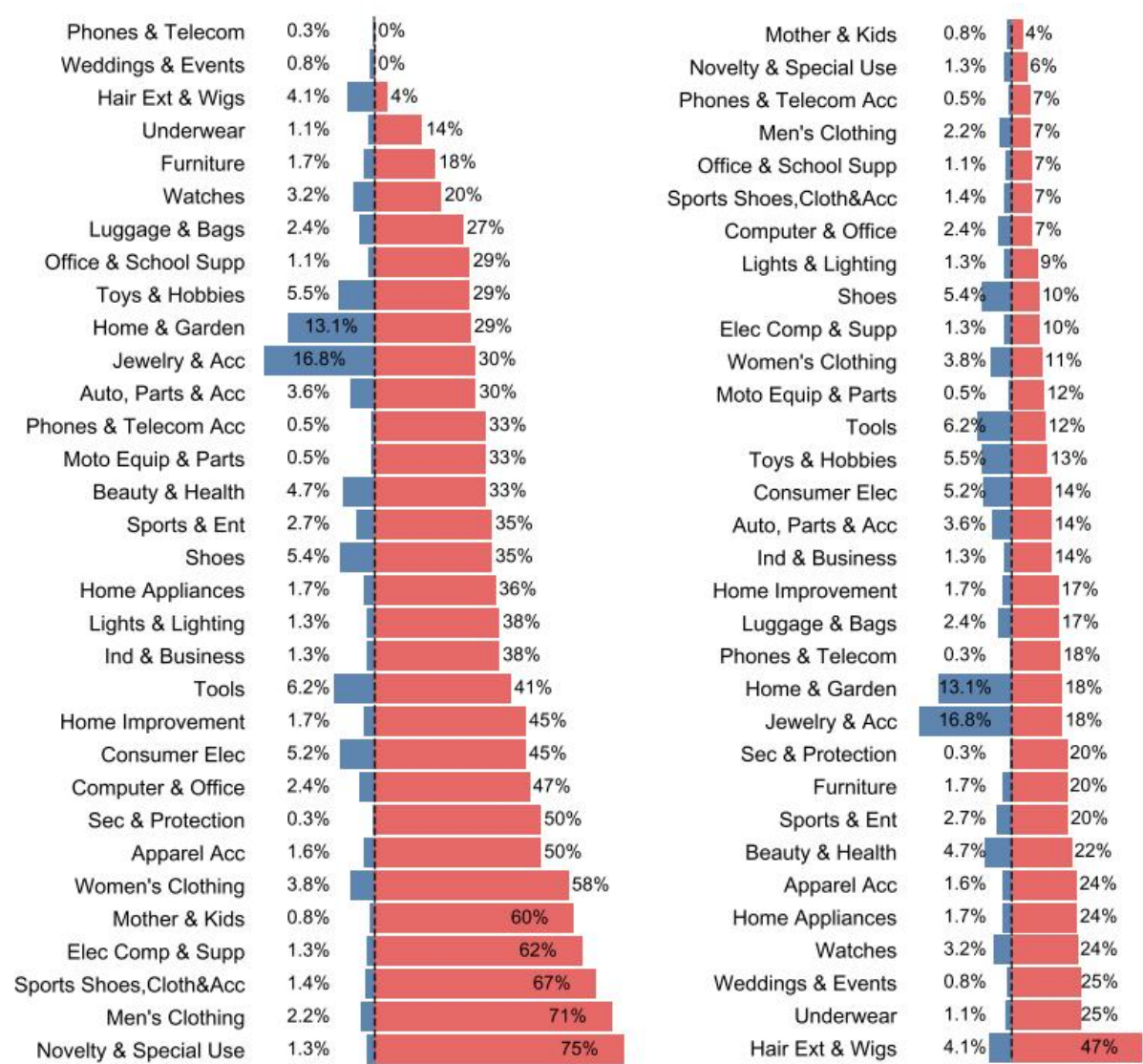


Figure 6: images/04086782116fff24e3be79008b46bacbc654261630c5dd80bb890283359aa7f9.jpg

Figure 5. Both figures show the category distribution on the left blue bars. Left: Challenging Product Distribution (CID). Right: Average Performance Distribution (APD).

图 5。两个图表都显示了左侧蓝色条形图中的类别分布。左侧：具有挑战性的产品分布 (CID)。右侧：平均性能分布 (APD)。

average performance stays below 18%. These findings underscore the challenges presented by HSCODECOMP, highlighting the need for more robust and generalizable approaches to HSCode prediction.

平均性能保持在 18% 以下。这些发现突显了 HSCODECOMP 提出的挑战，强调了需要更稳健且可推广的

方法来进行 HSCode 预测。

6.5. Test-time scaling Cannot Improve Performance Effectively

6.5. 测试时缩放无法有效提升性能